ILA — Intelligent Learning Assistant

Offline Model Selection Guide · 2026 Edition

7	6	3	0
Core LLM options	Embedding models	Hardware tiers	API calls needed

All models in this guide run fully offline using Ollama or llama.cpp — no internet connection, no API keys, no cloud dependency. Sections are organized by role in the ILA pipeline: LLMs for the tutoring brain, embedding models for RAG memory, and recommended stacks by hardware tier.

PART 1 — Core Language Models (LLMs)

The tutoring brain — generates explanations, answers questions, creates quizzes.

Phi-4-mini (3.8B) — Microsoft

VRAM
~3 GB

Apache 2.0 · ollama pull phi4-mini · 128K context window

Low VRAM
No GPU needed
Strong math/logic

Resource footprint (relative to 24 GB VRAM)

Advantages	Limitations
+ Runs on any modern laptop, no discrete GPU required	- Weaker broad factual recall vs larger models
+ Best-in-class math/logic for its size (80.4% MATH benchmark)	- Not optimal for long-form explanation generation
+ Remarkable 128K context window for a 3.8B model	- Reasoning degrades on complex multi-step problems
+ Fast inference — ideal for real-time quiz generation	- Smaller community fine-tune library vs Llama
+ Apache 2.0 license — fully commercial-safe	

Best for: Quiz generation, math tutoring, low-end hardware deployments

Llama 3.2 (3B) — Meta

VRAM
~2.5 GB

Llama 3 Community License · ollama pull llama3.2:3b · 128K context

Ultra low VRAM
Multilingual
Huge community

Resource footprint (relative to 24 GB VRAM)

Advantages	Limitations
+ Smallest viable instruction-following model	- Noticeably weaker reasoning vs Phi-4-mini at same size
+ Enormous community — many domain fine-tunes available	- Higher hallucination risk on factual questions
+ Excellent for bootstrapping and rapid prototyping	- Not ideal as production tutor for hard STEM subjects
+ Native multi-lingual support	- Commercial use requires Meta's license agreement

Best for: Prototyping, lightweight devices, multilingual content

Qwen3 (8B) — Alibaba

VRAM
~5 GB

Apache 2.0 · ollama pull qwen3:8b · Built-in thinking mode

Apache 2.0
Thinking mode
100+ languages

Resource footprint (relative to 24 GB VRAM)

Advantages	Limitations
+ Built-in thinking mode for step-by-step explanations — perfect for tutoring	- Thinking mode increases response latency significantly
+ Tops benchmarks in its size class for reasoning and coding	- Slightly verbose responses requiring prompt tuning
+ Apache 2.0 — no commercial restrictions	- Alibaba-origin model (irrelevant for fully offline use)
+ 100+ language support, excellent for Indian students	- 8B is strong but not as deep as 14B+ for hard science
+ Fits comfortably on 8 GB VRAM	

Best for: STEM tutoring, step-by-step explanations, multilingual ILA

Gemma 3 (12B) — Google

VRAM
~8 GB

Gemma Terms of Use · ollama pull gemma3:12b · 128K context

Strong instruction follow
128K context
Good RAG performance

Resource footprint (relative to 24 GB VRAM)

Advantages	Limitations
+ Excellent instruction following — great tutoring dialogue quality	- Google custom license — not Apache 2.0, check commercial terms
+ 128K context fits entire textbook chapters at once	- Heavier than Phi-4; dedicated GPU needed for good speed
+ Strong general knowledge vs smaller models	- Slightly behind Qwen3 at same size on benchmarks
+ Multimodal variant can process images in study materials	- Gemma 4 launched 2026; this line may become legacy quickly
+ Runs on RTX 3060 / 12 GB VRAM	

Best for: Balanced tutoring, good reasoning, 12 GB GPU setups

Mistral 7B Instruct v0.3 — Mistral AI

VRAM
~4.5 GB

Apache 2.0 · ollama pull mistral · Sliding window attention

Battle-tested
Fast inference
Proven RAG baseline

Resource footprint (relative to 24 GB VRAM)

Advantages	Limitations
+ Industry standard RAG baseline — most tutorials and examples target it	- Outperformed by Qwen3 8B on most current benchmarks
+ Very fast token generation, low latency tutoring experience	- No built-in reasoning / thinking mode
+ Huge fine-tune ecosystem across many domains	- Smaller training data cutoff vs newer models
+ Apache 2.0 — fully open for any use case	- Default 8K context; needs config changes for longer documents
+ Sliding window attention handles long contexts efficiently	

Best for: Fast prototyping, proven RAG pipelines, general conversational tutoring

DeepSeek-R1-Distill (14B) — DeepSeek

VRAM
~9–10 GB

MIT License · ollama pull deepseek-r1:14b · Chain-of-thought native

MIT License
Chain-of-thought
Top STEM reasoning

Resource footprint (relative to 24 GB VRAM)

Advantages	Limitations
+ Native chain-of-thought — shows working steps automatically	- Longer responses due to CoT traces — slower perceived speed
+ MIT license — most permissive of all options	- Reasoning traces consume significant context window
+ Best math and science reasoning per VRAM spent	- Less conversational than Qwen/Gemma for general chat
+ Distilled from 671B R1 model — far exceeds its apparent size	- Needs ~10–12 GB VRAM at Q4_K_M for smooth inference
+ Visible reasoning trace helps students understand solutions	

Best for: STEM adaptive tutoring, showing working, deep subject reasoning

Llama 3.3 (70B) — Meta

VRAM
~40–48 GB

Llama 3 License · ollama pull llama3.3:70b-instruct-q4_K_M · 128K context

Workstation only
Beats GPT-4 original
Highest quality

Resource footprint (relative to 24 GB VRAM)

Advantages	Limitations
+ Best overall quality — beats GPT-4 original on most benchmarks	- Requires 48 GB+ VRAM — dual RTX 3090/4090 or A100
+ Exceptional nuanced, adaptive explanations	- Very slow on consumer hardware (3–5 tok/s without high-end GPU)
+ 128K context can process an entire textbook at once	- High RAM makes multi-service ILA setup very challenging
+ Largest community and fine-tune ecosystem available	- Overkill for most tutoring tasks where smaller models suffice

Best for: Research labs, high-end servers, gold-standard tutoring quality

PART 2 — Embedding Models

Embedding models convert your PDFs and notes into vectors for semantic search (RAG retrieval). They run once at index time and are lightweight — all options below are fully offline.

BGE-M3 — BAAI

MIT License · 568M params · 8192 token max · 1000+ languages

Dense+sparse+multi-vec
1000+ languages
Top ILA pick

Pros	Cons
+ Handles short queries AND long textbook passages (8K tokens)	- 568M parameters — heavier than lightweight alternatives
+ Three retrieval modes in one model (dense, sparse, multi-vec)	- Slightly slower indexing speed vs smaller models
+ 1000+ language support — best for multilingual study material	
+ MIT license — fully free for any use	

Best for: Top recommendation for ILA — best overall for diverse study material

nomic-embed-text v1.5 — Nomic AI

Apache 2.0 · 137M params · 8192 token max · English focus

Apache 2.0
Matryoshka dims
Very fast

Pros	Cons
+ Fully open — weights, code, and training data all public	- English-dominant; weak on Indic and regional languages
+ Matryoshka embeddings: truncate dimensions to save storage	- Smaller model size means lower accuracy ceiling
+ Fastest of the group; ideal for real-time RAG	
+ 8K token context fits entire lecture notes	

Best for: Speed-critical pipelines, English-only content, storage efficiency

all-mpnet-base-v2 — sentence-transformers

Apache 2.0 · 110M params · 512 token max · English

Most downloaded HuggingFace
768-dim vectors
Battle-tested

Pros	Cons
+ Most downloaded embedding model on HuggingFace	- 512 token max input — requires aggressive chunking strategy
+ Trained on 1B+ sentence pairs — rich semantic capture	- Older model; newer alternatives surpass on benchmarks
+ 149+ fine-tuned subject-specific variants available	
+ Direct LangChain / LlamaIndex integration	

Best for: Quick setup, familiar RAG stacks, fine-tuning baseline

mxbai-embed-large-v1 — MixedBread AI

Apache 2.0 · 335M params · 512 token max · 1024-dim vectors

Top MTEB English
1024-dim
High accuracy

Pros	Cons
+ Among best English-only retrieval accuracy scores (MTEB)	- 512 token limit — requires aggressive chunking
+ High-dimensional vectors (1024) for fine semantic granularity	- English-only; not suitable for multilingual content
+ Apache 2.0 — no restrictions	- Poor cross-lingual retrieval scores (0.120 on CCKM benchmark)
+ Great for exam prep Q&A; retrieval tasks	

Best for: English study materials requiring high-precision retrieval

gte-multilingual-base — Alibaba / GTE

Apache 2.0 · Encoder-only architecture · Sparse + dense retrieval

Multilingual
Sparse+dense
10x faster than decoders

Pros	Cons
+ Encoder architecture — 10x inference speed vs decoder models	- Inconsistent performance across lower-resource languages
+ Generates both sparse and dense vectors in one pass	- Limited contrastive training data for some regional languages
+ Strong multilingual retrieval across Indian and global languages	
+ Very memory efficient for indexing large corpora	

Best for: Multilingual study materials, Hindi and regional language content

Nomic Embed Text V2 (MoE) — Nomic AI

Apache 2.0 · MoE architecture · 1.6B training pairs · ~100 languages

First MoE embed
1.6B training pairs
100 languages

Pros	Cons
+ First Mixture-of-Experts architecture for text embeddings	- Newer model — less production-tested than BGE-M3 or mpnet
+ Trained on 1.6B multilingual contrastive pairs	- MoE architecture adds deployment complexity
+ Strong semantic search, RAG, and recommendations	
+ Efficient at inference despite breadth of training	

Best for: Multilingual adaptive tutoring, diverse and mixed content types

PART 3 — Recommended Stacks by Hardware

Tier 1 — Budget / Laptop (≤8 GB RAM, no discrete GPU)

Minimal viable offline ILA

LLM	Phi-4-mini (3.8B) or Qwen3:4b
Embedding	nomic-embed-text v1.5
Runtime	Ollama + CPU inference
Vector DB	ChromaDB (local, persistent)
PDF parser	PyMuPDF
Framework	LangChain or LlamaIndex

Works on any 8 GB RAM laptop. ~3–5 tok/s inference speed. Supports full RAG + quiz generation pipeline.

Tier 2 — Mid-range (RTX 3060/4060 · 12–16 GB VRAM) ← Recommended

Recommended production ILA setup

LLM	DeepSeek-R1-Distill 14B or Qwen3 14B
Embedding	BGE-M3
Runtime	Ollama + CUDA acceleration
Vector DB	ChromaDB or Qdrant (local)
PDF parser	PyMuPDF + Unstructured
Adaptive ML	sklearn BayesianRidge / pyBKT / LightGBM

Strong STEM reasoning with visible CoT explanations. 15–25 tok/s GPU speed. Full adaptive quiz ML pipeline.

Tier 3 — High-end Workstation (RTX 4090 / dual GPU · 24–48 GB VRAM)

Research-grade ILA with maximum quality

LLM	Llama 3.3 70B or Qwen3 32B
Embedding	BGE-M3 + mxbai reranker
Runtime	vLLM + Ollama
Vector DB	Qdrant (persistent, full-featured)
PDF parser	Unstructured + OCR (pytesseract)
Adaptive ML	Full BKT + IRT (Item Response Theory) model

Near-frontier quality, full adaptive learning pipeline. High VRAM required.

All models are available via Ollama (ollama.com) or HuggingFace. VRAM estimates assume Q4_K_M quantization at 4K context. Actual usage varies with context length — longer contexts require significantly more memory. Guide compiled May 2026.